

SAICT iMohon

Project Understanding & Architecture Reference

Malaysian government (RTM) equipment request & distribution system · Generated 12 Aug 2026

A Malaysian government **equipment request and distribution management system** for RTM (Radio Televisyen Malaysia), deployed to `imohon.rtm.gov.my`. Staff submit equipment requests (*permohonan*) that pass through a multi-tier approval chain, then get distributed (*agihan*) with full delivery and acceptance tracking. All user-facing text is in **Bahasa Malaysia**; code identifiers are in English.

Stack & Shape

- **backend/** — Laravel 10 REST API (PHP 8.1+, MySQL). Sanctum token auth, Spatie Permission (6 roles) + Activity Log, nested-set categories & departments.
- **frontend/** — React 18 SPA (Create React App). Zustand state, React-Bootstrap 5, Axios with token interceptors.
- **Deployment** — `scripts/deploy.sh` (git pull → composer install → npm build) behind nginx + PHP 8.2-FPM.
- **Branches** — default dev branch is `dev`; work here is on `claude/project-understanding-sufy7s`.

The Two Core Workflows

1 · Permohonan (Request) `MOHON-{year}-{NNNN}`

User drafts a request → submits → **Manager** approves → **Admin** processes & finalizes. A 4-step state machine tracked on `MohonApproval` (`step` / `status`), mirrored onto `MohonRequest`.

2 · Agihan (Distribution) `AGIHAN-{year}-{NNNN}`

Admin creates a distribution request for an approved permohonan and assigns inventory items → **Boss** approves → Admin sets delivery info → User accepts delivery. Reference numbers are auto-generated in a model `booted()` hook.

MohonApproval state machine

step	status	Meaning	Actor
0	pending	Draft (not yet submitted)	User
1	pending	Submitted, awaiting Manager	Manager
2	approved / rejected	Manager decision	Manager
3	pending / approved	Admin processing	Admin
4	approved / rejected	Final decision, ready for distribution	Admin

Audit-trail design: approval records are *inserted*, never updated in place — each step is a new row. Backend logic lives in a `public static service layer` (`MohonService` , `MohonApprovalService` , distribution services); controllers are thin and role-grouped under `/api/{role}/...` , wired in `RouteServiceProvider.php` .

Practical Gotchas — Where Code Diverges from the Docs

These are the things worth knowing before touching the code — verified against the actual source, beyond what `CLAUDE.md` describes.

Area	Reality on the ground
Route-level RBAC (frontend)	There is none . <code>ProtectedRoute</code> only checks auth + email verification. Roles merely change which navbar links appear — any logged-in user can navigate directly to any admin/boss URL. Access control is enforced server-side per role prefix, not in the SPA.
DB foreign keys	Migrations have a pervasive typo — <code>->costrained()</code> instead of <code>->constrained()</code> . It is a silent no-op, so no FK constraints exist at the DB level ; relationships are enforced only in Eloquent.
Dead "Application" subsystem	<code>ApplicationService</code> , <code>DistributionService</code> , <code>DistributionApprovalService</code> , <code>DistributionAcceptanceService</code> , <code>User::applications()</code> , and <code>LoginTest</code> 's <code>/api/applications</code> all reference <code>Application</code> / <code>Distribution</code> models that don't exist . Leftovers from a prior design renamed to "Mohon" — treat as dead code.
Mass assignment	Domain models use <code>\$guarded</code> , not <code>\$fillable</code> (docs say fillable). Only <code>UserDepartment</code> declares an explicit <code>\$fillable</code> .
Tests	Pest 2, but domain coverage is essentially zero — only trivial example tests plus a broken <code>LoginTest</code> hitting a non-existent route. <code>RefreshDatabase</code> is commented out.
Frontend cruft	The <code>createResourceStore</code> factory is imported nowhere (dead) — every page hand-rolls the same store ~20x. Versioned duplicates remain (<code>Agihan</code> vs <code>Agihan-v2</code> , <code>MohonDistributionItem</code> vs <code>-v2</code>). App is hard-gated behind

`REACT_APP_MODE==='production'` and a 768px **desktop-only** block. Router lives entirely in `index.js` (no `App.js`).

Minor

Duplicate admin distribution routes; two migrations share timestamp `2026_03_04_042748`; the `global.php` route group lacks `auth:sanctum` at the group level.

Key Models & Relationships

- `User` → `UserProfile` → `UserDepartment` (nested set). Roles via Spatie: `user` `manager` `admin` `boss` `system` `guest`
- `MohonRequest` → many `MohonItem` (→ `Category`), many `MohonApproval` (ASC), many `MohonDistributionRequest` (DESC).
- `MohonDistributionRequest` → many `MohonDistributionApproval` (→ `boss`), many `MohonDistributionItem`.
- `MohonDistributionItem` → `Inventory`, `Category`, `MohonItem`; hasOne `Delivery` + hasOne `Acceptance`.

Source of truth: this reference reflects the state of the repository as read on 12 Aug 2026 on branch `claude/project-understanding-sufy7s`. The high-level workflow diagrams and state tables match `CLAUDE.md`; the "Practical Gotchas" section captures where the live code diverges from or carries legacy baggage relative to that document.